

PROJECT REPORT

EXERCISE NO – 09

Project title:

**WORKER HEALTH AND SAFETY MONITORING SYSTEM
FOR CONSTRUCTION INDUSTRIES**

Submitted by :

Arundepak S – 24MER004

Deepak P – 24MER006

Dharanidharan R S – 24MER007

Dinesh K – 24MER009

Vishnivarthan P – 24MER063

1. Project Overview

The Worker Health and Safety Monitoring System is an IoT-enabled system developed to continuously observe the health and environmental safety conditions of construction workers. An ESP8266 NodeMCU is used as the primary controller, which receives data from a Pulse Sensor, DHT11 temperature and humidity sensor, MQ2 gas sensor, and NEO-6M GPS module.

The Pulse Sensor is used to monitor the worker's heart rate, whereas the DHT11 sensor measures the surrounding temperature and humidity. The MQ2 sensor identifies the presence of smoke or combustible gases and provides a gas-alert indication. The NEO-6M GPS module supplies the worker's latitude, longitude, and movement-related information for location tracking.

The ESP8266 collects and processes the readings from all the sensors and establishes an Internet connection through Wi-Fi. The measured data is sent to ThingSpeak Cloud, where the parameters can be viewed using numerical values, charts, and graphs. When conditions such as high temperature, gas detection, or abnormal heart-rate values are identified, the system indicates an UNSAFE condition.

2. Problem Statement

Construction workers are regularly exposed to several workplace hazards, including extreme temperatures, harmful gases, smoke, and physically demanding activities. Conventional safety monitoring techniques may not continuously provide information about a worker's health condition or location, particularly during emergency situations.

Hence, there is a requirement for an IoT-based real-time monitoring system capable of observing essential health and environmental parameters and detecting potentially dangerous conditions. The proposed system integrates an ESP8266, Pulse Sensor, DHT11, MQ2 gas sensor, and NEO-6M GPS module to collect safety-related information and transmit it to ThingSpeak Cloud for remote monitoring and analysis.

The system provides early alerts when abnormal conditions are identified, helping supervisors respond quickly and improving the overall safety and protection of construction workers.

3. Proposed Solution

The proposed solution is an IoT-based Worker Health and Safety Monitoring System developed to continuously track the health condition, environmental parameters, and geographical location of construction workers in real time.

The system employs an ESP8266 NodeMCU as the main processing unit. The Pulse Sensor measures the worker's heart rate, while the DHT11 sensor monitors ambient temperature and humidity. The MQ2 gas sensor detects smoke and combustible gases present in the working environment. The NEO-6M GPS module is used to determine the worker's geographical position through latitude and longitude information.

The ESP8266 receives and processes the sensor data and connects to the Internet using Wi-Fi. The collected information is uploaded to ThingSpeak Cloud, allowing supervisors to remotely observe the worker's health and safety parameters through a dashboard.

When conditions such as gas detection, excessive temperature, or abnormal heart-rate conditions occur, the system classifies the situation as UNSAFE. This gives an early indication of hazardous conditions and enables supervisors to take suitable safety measures.

4. System Architecture

- **Worker Monitoring Layer:** The monitoring unit is positioned with or close to the construction worker to continuously observe health and environmental parameters.
- **Pulse Monitoring:** The Pulse Sensor captures the worker's pulse signal, which is processed by the ESP8266 to determine the heart-rate condition.
- **Environmental Monitoring:** The DHT11 sensor measures the surrounding temperature and humidity, while the MQ2 sensor identifies smoke and combustible gas conditions.
- **Location Monitoring:** The NEO-6M GPS module provides latitude, longitude, speed, and satellite information for determining the worker's geographical location.
- **Processing Unit:** The ESP8266 NodeMCU functions as the central controller, collecting sensor readings and comparing them with predefined safety thresholds.
- **Safety Assessment:** The ESP8266 evaluates the collected information and determines whether the condition is SAFE or UNSAFE, based on parameters such as abnormal heart rate, high temperature, or gas detection.
- **Wireless Communication:** The ESP8266 uses Wi-Fi connectivity to transfer the collected sensor information to the cloud.
- **Cloud Monitoring:** ThingSpeak Cloud receives and stores the sensor readings from the ESP8266 and presents them through numerical values, charts, and graphs.
- **Remote Monitoring:** Supervisors can remotely observe the worker's health parameters, environmental conditions, GPS location, and overall safety status through the ThingSpeak dashboard.
- **Data Analysis and Safety Response:** Stored historical data can be analyzed to identify hazardous patterns and support faster decision-making and appropriate safety actions.

5. Circuit Table

Component	Pin	Microcontroller Pin (ESP8266/ESP32)
Pulse Sensor	VCC	3.3V
Pulse Sensor	SIGNAL	A0
Pulse Sensor	GND	GND
DHT11	DATA	GPIO 4
DHT11	VCC	5V
DHT11	GND	GND
MQ2 Gas Sensor	D0	GPIO 5
MQ2 Gas Sensor	VCC	5V
MQ2 Gas Sensor	GND	GND
NEO-6M GPS	TX	GPIO 14
NEO-6M GPS	RX	GPIO 12
NEO-6M GPS	VCC	3.3V

6. Circuit Design

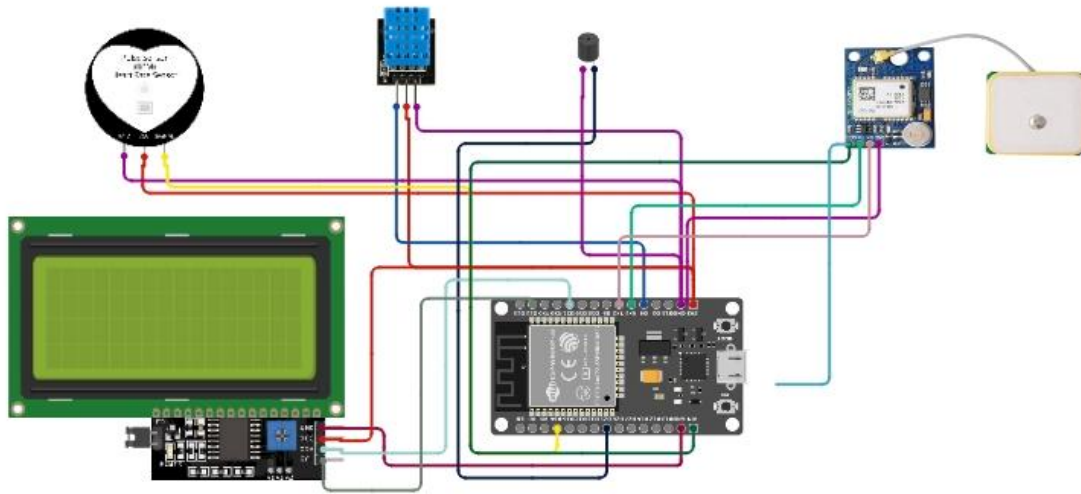


Fig. 1 Circuit wiring diagram

7. Algorithm

- **Start and Initialization:** Switch on the ESP8266 and initialize the Pulse Sensor, DHT11, MQ2 sensor, and NEO-6M GPS module.
- **Wi-Fi Connection:** Establish a connection between the ESP8266 and the configured Wi-Fi network for Internet and cloud communication.
- **Sensor Data Acquisition:** Obtain the worker's heart rate, temperature, humidity, gas detection status, and GPS location information.
- **Data Processing:** The ESP8266 processes the sensor readings and compares them against the predefined safety thresholds.
- **Safety Evaluation:** If an abnormal heart rate, excessive temperature, or gas detection is identified, the system sets the condition to UNSAFE. Otherwise, the condition is considered SAFE.
- **GPS Monitoring:** Receive valid latitude, longitude, speed, and satellite information from the NEO-6M GPS module.
- **Cloud Upload:** Transmit the collected sensor parameters and safety status to the respective ThingSpeak fields through Wi-Fi.
- **Display and Repeat:** Display the current sensor readings on the Serial Monitor and continuously repeat the monitoring and cloud-upload process.

8. CODING

```
#include <ESP8266WiFi.h>
#include <ESP8266HTTPClient.h>
#include <WiFiClient.h>
#include <DHT.h>
#include <TinyGPSPlus.h>
#include <SoftwareSerial.h>

const char* WIFI_SSID = "SK2509";
const char* WIFI_PASSWORD = "SRDS2506";
const char* THINGSPEAK_API_KEY = "5U30FAOS0SXE1ET9";
const char* THINGSPEAK_SERVER = "http://api.thingspeak.com/update";

#define DHT_PIN 4
#define DHT_TYPE DHT11
DHT dht(DHT_PIN, DHT_TYPE);

#define PULSE_PIN A0
#define MQ2_PIN 5
#define GPS_RX_PIN 14
#define GPS_TX_PIN 12

SoftwareSerial gpsSerial(GPS_RX_PIN, GPS_TX_PIN);
TinyGPSPlus gps;

float temperature = 0.0;
float humidity = 0

int pulseValue = 0;
int heartRate = 0; int
gasStatus = 0; float
latitude = 0.0;

float longitude = 0.0;
float gpsSpeed = 0.0;
bool gpsFix = false;
unsigned long lastSensorRead = 0;
unsigned long lastDisplay = 0;
unsigned long lastThingSpeak = 0;

const unsigned long SENSOR_INTERVAL = 1000;
const unsigned long DISPLAY_INTERVAL = 2000;

const float MAX_TEMPERATURE = 40.0;
```

```
const int MIN_HEART_RATE = 50;
const int MAX_HEART_RATE = 120;
void setup()
{
  Serial.begin(115200);
  delay(1000);
  Serial.println();
  Serial.println("=====");
  Serial.println(" WORKER HEALTH & SAFETY MONITORING SYSTEM");
  Serial.println("=====");
  pinMode(PULSE_PIN, INPUT);
  pinMode(MQ2_PIN, INPUT);
  gpsSerial.begin(9600);
  Serial.println();
  Serial.println("GPS CONFIGURATION");
  Serial.println("-----");
  Serial.println("NEO-6M TX -> D5 / GPIO14");
  Serial.println("NEO-6M RX -> D6 / GPIO12");
  Serial.println("GPS Baud -> 9600");

  Serial.println();
  Serial.println("GPS started...");
  Serial.println("Waiting for GPS data...");
  Serial.println();
  Serial.print("Connecting to WiFi");
  WiFi.begin(WIFI_SSID, WIFI_PASSWORD);
  while (WiFi.status() != WL_CONNECTED)

  {
    readGPS();
    delay(300);
    Serial.print(".");
  }
```

```
Serial.println();
Serial.println("WiFi Connected!");
Serial.print("IP Address: ");
Serial.println(WiFi.localIP());
Serial.println();
Serial.println("SYSTEM READY");
Serial.println();
}

void readGPS()
{
  while (gpsSerial.available() > 0)
  {
    char c = gpsSerial.read();
    gps.encode(c);
  }
  if (gps.location.isValid())
  {
    latitude = gps.location.lat();
    longitude = gps.location.lng();
    gpsFix = true;
  }
  else
  {
    gpsFix = false;
  }
  if (gps.speed.isValid())
  {
    gpsSpeed = gps.speed.kmph();
  }
}

void readDHT()
{
  float t = dht.readTemperature();
```

```
float h = dht.readHumidity();
if (!isnan(t))
{
    temperature = t;
}
if (!isnan(h))
{
    humidity = h;
}
}
void readMQ2()
{
    int state = digitalRead(MQ2_PIN);
    if (state == LOW)
    {
        gasStatus = 1;
    }
    else
    {
        gasStatus = 0;
    }
}
void readPulse()
{
    pulseValue = analogRead(PULSE_PIN);
    if (pulseValue > 550)
    {
        heartRate = 80;
    }
    else
    {
        heartRate = 0;
    }
}
```



```

bool isUnsafe()
{
    if (gasStatus == 1)
    {
        return true;
    }
    if (temperature > MAX_TEMPERATURE)
    {
        return true;
    }
}

```

9. System Working

- The ESP8266 NodeMCU operates as the central controller and initializes the connected sensors along with the Wi-Fi connection.
- The Pulse Sensor detects the worker's pulse and provides the corresponding signal to the ESP8266 through the A0 analog input.
- The DHT11 sensor measures the surrounding temperature and humidity to identify potentially uncomfortable or hazardous environmental conditions.
- The MQ2 sensor is connected through its digital output and detects smoke or combustible gases, producing a corresponding gas detection status.
- The NEO-6M GPS module receives satellite signals and provides the worker's latitude, longitude, speed, and GPS status.
- The ESP8266 processes the information received from all sensors and compares the values with predefined safety thresholds.
- If high temperature, gas detection, or abnormal heart-rate conditions are detected, the system classifies the worker's condition as UNSAFE. Otherwise, the system displays SAFE.
- Through Wi-Fi connectivity, the ESP8266 sends the collected information to ThingSpeak Cloud.
- ThingSpeak stores and displays the worker's health parameters, environmental readings, GPS information, and safety condition using numerical displays and graphical representations.
- The monitoring process continues repeatedly, enabling real-time remote observation of worker safety conditions.

10. Bill of Materials

S.No	Component	Quantity
1	ESP8266 NodeMCU	1
2	Pulse Sensor	1
3	DHT11 Sensor	1
4	MQ2 Gas Sensor	1
5	NEO-6M GPS Module	1
6	Breadboard	1
7	USB Cable	1
8	ThingSpeak Account	1
9	Wi-Fi Network	1
10	Jumper Wires	Required

11. Prototype Implementation

- **Hardware Assembly:** The ESP8266 NodeMCU is mounted on a breadboard and connected to the Pulse Sensor, DHT11, MQ2 gas sensor, and NEO-6M GPS module.
- **Sensor Integration:** The Pulse Sensor is connected to A0, MQ2 to D1, DHT11 to D2, and the NEO-6M GPS TX/RX lines are connected to D5/D6 respectively.
- **Programming:** The system is programmed using the Arduino IDE with the required libraries for DHT11, TinyGPSPlus, SoftwareSerial, Wi-Fi, and ThingSpeak communication.
- **Wi-Fi and Cloud Configuration:** The system is configured with the required Wi-Fi credentials and ThingSpeak Write API Key. Eight ThingSpeak fields are created for heart rate, temperature, humidity, gas status, latitude, longitude, GPS speed, and safety status.
- **Sensor Data Collection:** The prototype continuously acquires health, environmental, and location-related information from the connected sensors.
- **Safety Evaluation:** The collected readings are compared with predefined safety limits to determine the SAFE or UNSAFE condition.
- **GPS Tracking:** Latitude, longitude, and speed information are recorded whenever a valid GPS satellite fix is obtained.
- **Cloud Monitoring:** The processed sensor information is periodically transmitted to ThingSpeak through the Wi-Fi connection.
- **Testing:** The prototype is tested under normal operating conditions and simulated unsafe situations, such as increased temperature and gas detection.

Continuous Operation: The ThingSpeak dashboard is repeatedly updated to support continuous monitoring of worker safety conditions

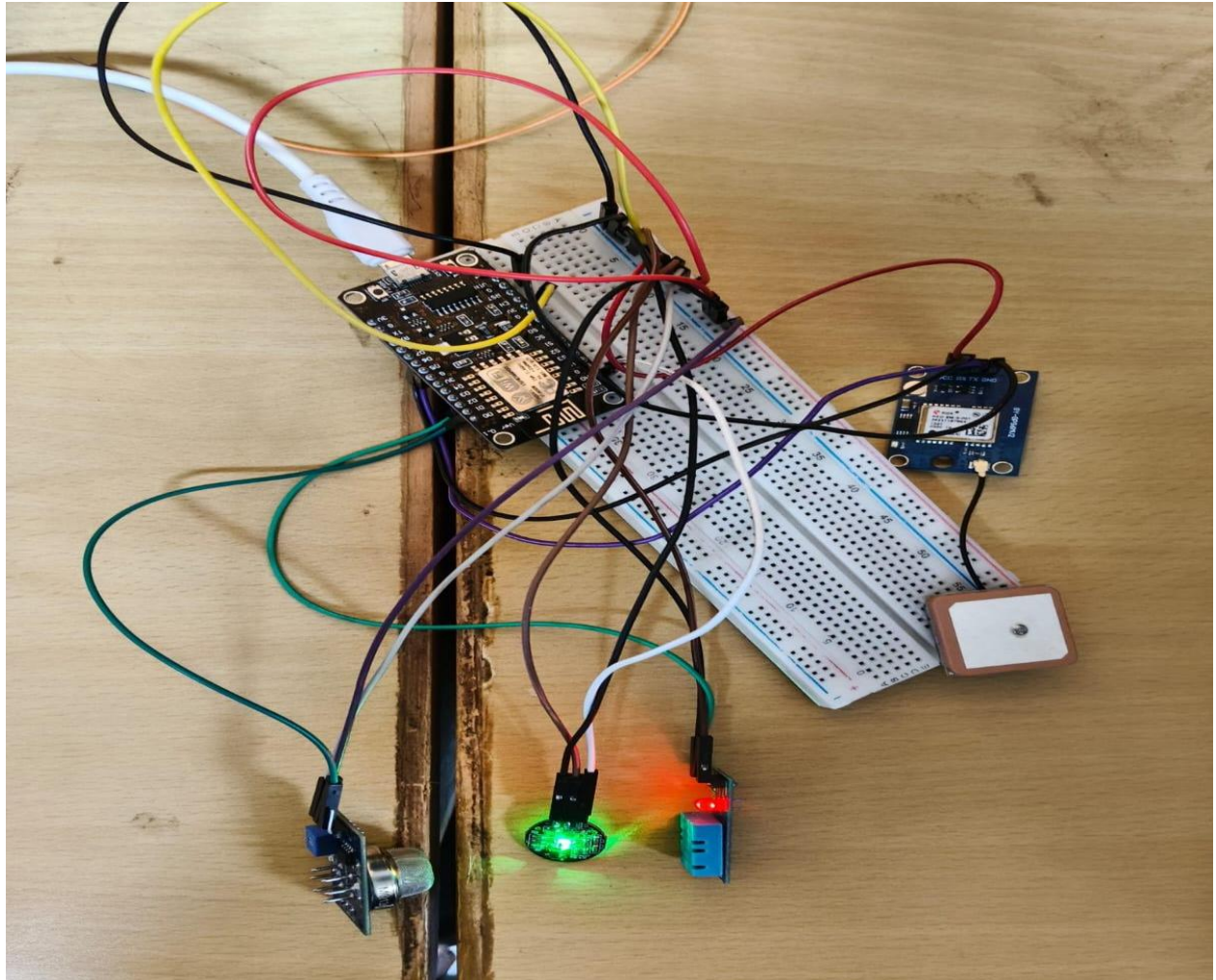


Fig. 2 Assembled prototype — Pulse Sensor, DHT11, MQ2, NEO-6M GPS wired to ESP8266

12. Results & Performance Analysis

- The developed prototype successfully monitors the worker's heart rate, temperature, humidity, gas status, and geographical location using the ESP8266.
- The Pulse Sensor provides pulse-related information that can be used to identify abnormal heart-rate conditions.
- The DHT11 sensor effectively measures the surrounding temperature and humidity, helping in the identification of uncomfortable or potentially unsafe environmental conditions.
- The MQ2 sensor detects smoke or gas through its digital output and provides a Safe/Gas Detected indication.
- The NEO-6M GPS module provides latitude, longitude, speed, and satellite information when a valid GPS fix is obtained.
- The ESP8266 processes the sensor information and determines the overall SAFE or UNSAFE condition based on the programmed limits.
- The collected sensor information is successfully transmitted to ThingSpeak Cloud through Wi-Fi and displayed using different fields, charts, and graphs.
-

- The system enables continuous remote monitoring through the ThingSpeak dashboard.
- The prototype demonstrates that integrating health monitoring, environmental sensing, GPS tracking, and cloud connectivity can provide an effective approach for improving construction worker safety.
- The developed system shows good potential for real-time safety assessment and early warning generation, subject to further sensor calibration and industrial-level validation before practical deployment.

```

SAFETY STATUS      :  SAFE
=====

=====
                                WORKER SAFETY DATA
=====
Pulse ADC          :  567
Heart Rate         :  80 BPM
Temperature        :  30.9 C
Humidity           :  59.2 %
Gas Status         :  SAFE

GPS INFORMATION
-----
GPS Characters     :  99691
GPS Sentences      :  32
GPS FIX           :  YES
Latitude          :  11.270588
Longitude         :  77.603981
GPS Speed         :  1.80 km/h
Satellites        :  10
HDOP              :  1.10
Altitude          :  88.50 m
GPS Date          :  19/08/2026
GPS Time          :  08:30:09

SAFETY STATUS      :  SAFE
=====

```

Fig. 3 Serial Monitor — live sensor readings (heart rate, temp, humidity, gas, GPS) and ThingSpeak upload log



Fig. 4 ThingSpeak dashboard — live data with SAFE status

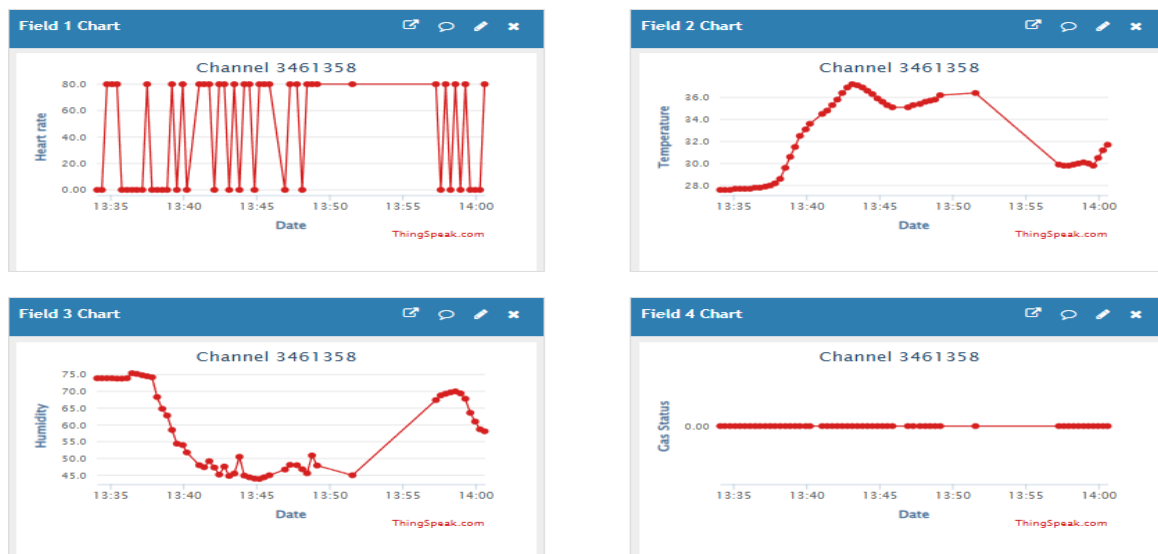


Fig. 5 ThingSpeak dashboard — live data with UNSAFE status (gas/temp/heart-rate alert)

